A Leakage-Aware Benchmark for Ultra-Small Language Models on Structured Speech-Command Routing

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Abstract. *Motivated by edge and on-device deployment, structured command routing benefits from ultra-small language models ($\leq$ 3B parameters). We present a reproducible benchmark evaluating five sub-3B instruction-tuned models across four families on a transcript-to-JSON routing task derived from Fluent Speech Commands (FSC). We evaluate under an explicit two-tool contract with balanced policy abstention (50% calls, 50% abstentions), comparing the official speaker-disjoint split ($n = 1{,}934$) against an audited phrase-disjoint partition ($n = 2{,}296$, 38 template clusters). Under the strict contract, a transparent lexical baseline achieves 77.40% exact match, outperforming every zero-shot neural model. Qwen3.5-2B attains 100% validity but collapses to universal abstention on phrase-disjoint test ($R_{call}^{exact} = 0\%$), while three models score 0% contract validity due to schema deviations. Supervised adaptation (LoRA) provides proof-of-concept contract learnability, establishing guidelines for edge SLM evaluation.*

1. Introduction and Scope

Motivated by edge and on-device deployment scenarios, we benchmark ultra-small language models (SLMs, $\leq$ 3B parameters) for natural language command routing. Evaluating these models zero-shot introduces two pitfalls: (i) template leakage across speakers in benchmark splits, and (ii) degenerate policy compliance, where models achieve accuracy by trivial abstention.

We evaluate five checkpoints in the sub-3B regime [4] (0.5B–2B parameters): Qwen2.5-0.5B, Qwen3.5-0.8B/2B [7], TinyLlama-1.1B [5], and SmolLM2-1.7B [6]. The input is a transcription from Fluent Speech Commands (FSC) [1]. The output must be a validated JSON object with three required root keys: `action`, `tool`, and `arguments`. Valid calls use `media_control` (`action`$\in${`play`, `pause`}) or `volume_adjust` (`direction`$\in${`up`, `down`}); abstention uses `action`:`"abstain"`, `tool`:`null`, `arguments`:{}. All other 27 native FSC intents map to policy abstention. Across all splits, the derived dataset totals 14,956 records (7,478 calls, 7,478 abstentions). We evaluate on FSC's official speaker-disjoint test ($n = 1{,}934$, 967 calls + 967 abstentions, derived from FSC's 3,793 test audios) and an audited phrase-disjoint test ($n = 2{,}296$, 1,148 calls + 1,148 abstentions, 38 template clusters; not speaker-disjoint by design).

2. Experimental Results

As shown in Table 1 and Figure 1, the lexical control dominates all zero-shot models. Exact Call Recall (R_{call}^{exact}) denotes the ratio of correct calls with exact tool and arguments over gold call instances. Strict Template Exact denotes the proportion of template clusters

Table 1. Phrase-disjoint vs. Official test results (%, JSON = parseable JSON).

	Phrase-Disjoint (Unseen Templates)				Official (Speaker Split)		
System	JSON	Valid	Exact	R_{call}^{exact}	Valid	Exact	Template Exact
Always abstain	100.00	100.00	50.00	0.00	100.00	50.00	75.71
Lexical control	100.00	100.00	**77.40**	**54.79**	100.00	**77.92**	**89.07**
Qwen2.5-0.5B	70.17	0.00	0.00	0.00	0.00	0.00	0.00
Qwen3.5-0.8B	100.00	60.15	43.60	33.45	73.47	54.65	56.28
TinyLlama-1.1B	43.42	0.00	0.00	0.00	0.00	0.00	0.00
SmolLM2-1.7B	96.47	0.00	0.00	0.00	0.78	0.00	0.00
Qwen3.5-2B	100.00	100.00	50.00	0.00	100.00	50.98	76.11

where 100% of items are correctly routed. On phrase-disjoint test, Qwen3.5-2B outputs `abstain` on 100% of inputs ($R_{call}^{exact} = 0\%$, Exact 50.00%); on official test, it emits 19 correct calls ($R_{call}^{exact} = 1.96\%$) reaching 50.98% exact match and 76.11% Template Exact. Qwen3.5-0.8B is the only model with substantial call coverage, but exhibits an observed split gap of 11.05 pp from official (54.65%) to phrase-disjoint (43.60%). Because the phrase-disjoint cluster bootstrap 95% CI is wide ([26.02%, 63.35%], over 38 clusters with 9 call and 29 abstain clusters), this gap is descriptive rather than a formal causal test.

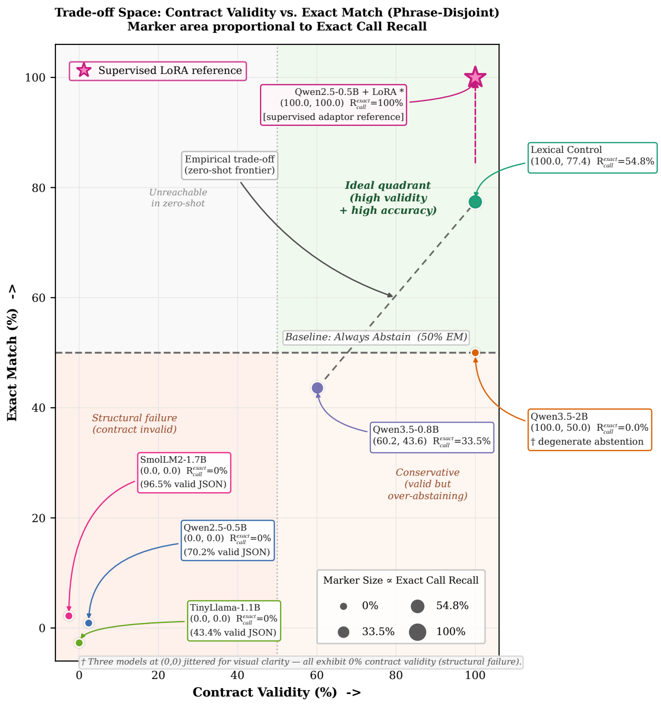

Figure 1. Trade-off space on phrase-disjoint test. Marker area $\propto$ exact call recall. The lexical baseline dominates zero-shot SLMs; Qwen3.5-2B collapses to 100% abstention on phrase-disjoint. Star: supervised in-distribution reference ($n = 1{,}934$).

Figure 2 reveals the exact point of structural failure: SmolLM2-1.7B produces 96.47% syntactically parseable JSON, but 0% valid contract objects due to schema deviations (missing action/tool keys or invalid parameters). Unconstrained zero-shot decoding is insufficient for sub-3B models to adhere to rigid task schemas.

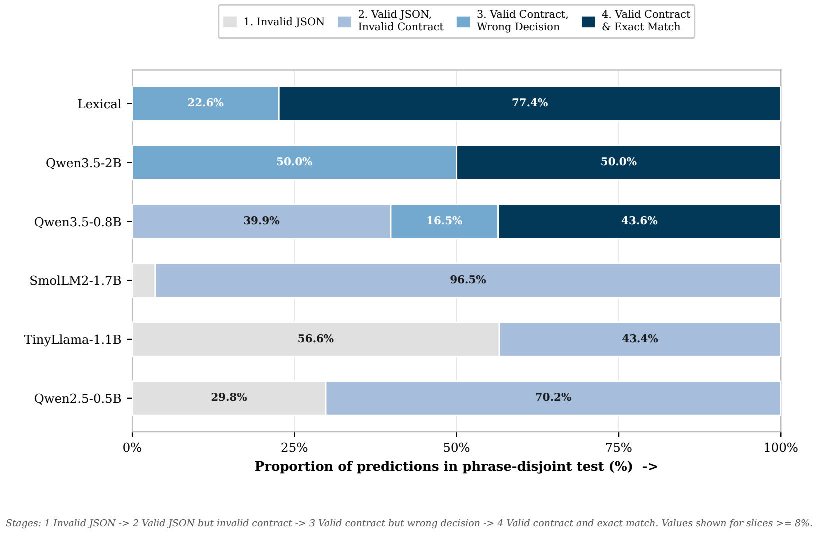

Figure 2. Structural cascade decomposing outputs into: (1) invalid JSON, (2) valid JSON but invalid schema, (3) valid contract but wrong call, and (4) exact match.

3. Discussion, Adaptation, and Conclusions

Two orthogonal techniques address this bottleneck: (i) **Grammar-Constrained Decoding** (e.g., GBNF in `llama.cpp`, Outlines [9]), which forces syntactic compliance by construction (collapsing stages 1–2 of Figure 2 to zero, though semantic discrimination still relies on the model); and (ii) **Supervised LoRA Adaptation** [8] as a proof of concept: a Qwen2.5-0.5B adapter ($r = 16, \alpha = 32$, lr 2×10^{-4}, 2 epochs, effective batch 16, RTX 5090) trained on official train reaches 100% validity and 100% exact on official test ($n = 1{,}934$), showing the contract is learnable, not that zero-shot routing is solved. The phrase-disjoint split uses NFKC normalization, whitespace collapsing, punctuation removal except apostrophes, and SHA-256 intent-stratified hashing (seed 20260830, target 70/15/15 split, 38 test clusters; not speaker-disjoint by design).

In conclusion, zero-shot structured routing without grammar guidance is fragile in the ultra-small regime. Benchmarks must decouple syntactic validity from semantic utility, audit template leakage, and incorporate transparent non-neural baselines. Extending this evaluation to multi-domain SLU benchmarks like SLURP [2] and compositional parsing corpora like STOP [3] represents an essential next step.

References

[1] L. Lugosch et al. Speech model pre-training for spoken language understanding. *Interspeech*, 2019.
[2] E. Bastianelli et al. SLURP: A Spoken Language Understanding Resource Package. *EMNLP*, 2020.
[3] P. Tomasello et al. STOP: A dataset for spoken task-oriented semantic parsing. *IEEE SLT*, 2023.
[4] Z. Liu et al. MobileLLM: Optimizing sub-billion parameter models for on-device. *ICML*, 2024.
[5] P. Zhang et al. TinyLlama: An open-source small language model. *arXiv:2401.02385*, 2024.
[6] L. Lozhkov et al. SmolLM2: When Smol Goes Big. *arXiv:2502.02737*, 2025.
[7] Qwen Team. Qwen3.5 model card. *Hugging Face*, 2026.
[8] E. J. Hu et al. LoRA: Low-rank adaptation of large language models. *ICLR*, 2022.
[9] B. T. Willard and R. Louf. Efficient guided generation for large language models. *arXiv:2307.09702*, 2023.